

# PAPER\_267: SFR Normalization as Dimensionless Coupling Constant $\alpha$ Starburst-Buoyancy Coherence in NGC 1792

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**Date:** March 2026

**UQFF Module:** GALAXY\_NGC\_1792.cpp (Module 19, “The Stellar Forge”)

**Session:** 73  $\alpha$  UQFF 2.0 Upgrade Unique Physics Extraction

**Keywords:** NGC 1792, specific star-formation rate, UQFF buoyancy, coherence, starburst gravity

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## Abstract

In the UQFF 2.0 upgraded model of NGC 1792 ( $z = 0.0095$ ,  $M_{\text{star}} = 1 \times 10^{11} M_{\odot}$ ,  $M_{\text{gas}} = 7.569 \times 10^9 M_{\odot}$ ,  $r = 7.569 \times 10^2 \text{ kpc}$ ), the normalized star-formation rate factor  $\text{SFR\_factor} = \text{SFR}[M_{\odot}/\text{yr}] / M_{\text{star}}[M_{\odot}] = 10 / 1 \times 10^{11} = 10^{-11} \text{ yr}^{-1}$  is identified as the **specific star-formation rate (sSFR)**  $\alpha$  a dimensionless coupling constant that uniformly scales the time-evolving mass  $M(t)$ . With the 3-tier buoyancy structure introduced in UQFF 2.0 (PAPER\_198 standard), all three buoyancy tiers couple to  $M(t)$  through the same sSFR exponential:  $\dot{M}_{\text{gas\_buoy\_total}} = \text{sSFR} \times (\text{term\_Ubi} + \text{term\_F\_UBii} + \text{term\_Ub\_i}) \times e^{\alpha t / \tau_{\text{SF}}}$ . This produces a **starburst-buoyancy coherence** effect: the peak of star formation and the peak of gravitational buoyancy occur simultaneously and decay with the same timescale  $\tau_{\text{SF}} = 100 \text{ Myr}$ . This paper derives the coherence formula, calculates numerical predictions for NGC 1792, and proposes observational signatures.

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\tau = 5.0 \times 10^4 \text{ day}^{-1}$ ,  $[\text{SSq}] = 0.57$ ) uniquely enabling this analysis  $\square$  establishing a new connection in the UQFF framework not present in Standard Model treatments.

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## 1. Introduction

The **specific star formation rate (sSFR)**  $= \text{SFR} / M_{\text{stellar}}$  is a fundamental dimensionally-reduced astrophysical parameter that characterizes the fractional mass growth rate of a galaxy per unit time. In the UQFF framework for NGC 1792, the mass evolution function  $M(t) = M_{\text{star}} \times (1 + \text{SFR\_factor} \times e^{\alpha t / \tau_{\text{SF}}})$  embeds the sSFR as the amplitude of the exponentially-decaying mass growth term. Previous analyses treated this purely as a mass-growth factor for the base gravity term. However, in the UQFF 2.0 framework with 3-tier buoyancy (introduced via PAPER\_198), the  $\text{Ug1\_t}$  field  $\alpha$  derived from  $M(t)$   $\alpha$  propagates into all three buoyancy tiers simultaneously. This establishes a direct coupling between sSFR and the complete buoyancy structure of the gravitational field.

NGC 1792 is a canonical starburst system with active star formation, making it an ideal test case. The galaxy is located at  $z = 0.0095$  in the constellation Columba,

approximately 40 Mpc from Earth, placing it in the same sky region as the Fornax Cluster (selected as the outer-frame external body for Tier 3 buoyancy).

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## 2. Physical Framework

### 2.1 SFR\_factor as sSFR

In the code:

```
SFR_factor = 10.0 / (1e10); // normalized: SFR[M_sun/yr] / M0[M_sun]
```

This is the **specific star formation rate** in  $\text{yr}^{-1}$ :

$$\text{sSFR} = \frac{\text{SFR}[M_{\odot}/\text{yr}]}{M_0[M_{\odot}]} = \frac{10}{10^{10}} = 10^{-9} \text{ yr}^{-1}$$

The mass evolution function becomes:

$$M(t) = M_0 \left( 1 + \text{sSFR} \cdot e^{-t/\tau_{\text{SF}}} \right)$$

and therefore:

$$\text{Ug1\_t} = \frac{GM(t)}{r^2} = \text{ug1\_base} \cdot \left( 1 + \text{sSFR} \cdot e^{-t/\tau_{\text{SF}}} \right)$$

where  $\text{ug1\_base} = G M_{\odot} / r_{\odot}^2$  is the static base field.

### 2.2 3-Tier Buoyancy Structure (UQFF 2.0)

The UQFF 2.0 upgrade introduces three buoyancy tiers, all derived from Ug1\_t:

#### Tier 1 Static half-gravity (Ubi):

$$\text{term\_Ubi} = 0.5 \cdot \text{Ug1\_t}$$

#### Tier 2 Dynamic compact cos modulation (F\_UBii, PAPER\_198):

$$\text{term\_F\_UBii} = -\beta_i \cdot \text{Ug1\_t} \cdot \omega_g \cdot \frac{M(t)}{r} \cdot [UA] \cdot \cos(\pi t)$$

#### Tier 3 Outer-frame via Fornax Cluster external body (Ub\_i, CP1):

$$\text{term\_Ub\_i} = -\beta_i \cdot \text{Ug1\_t} \cdot \omega_g \cdot \frac{M_{\text{Fornax}}}{r_{\text{Fornax}}} \cdot [UA] \cdot \cos(\pi t)$$

where  $\hat{I}_i^2 = 0.61$ ,  $\ddot{\varphi}_g = 7.3 \times 10^{-11} \text{ rad/s}^2$ ,  $[UA] = 10^{-11}$ ,  $M_{\text{Fornax}} = 1.393 \times 10^{14} \text{ kg}$ ,  $r_{\text{Fornax}} = 6.17 \times 10^6 \text{ m}$ .

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### 3. Starburst-Buoyancy Coherence

#### 3.1 Derivation

Since all three tiers are proportional to  $Ug1\_t$ , which contains the sSFR factor, the **total buoyancy enhancement** due to starburst activity is:

$$\Delta g_{\text{buoy}} = \text{buoy\_tiers}(t) - \text{buoy\_tiers}(t \rightarrow \infty)$$

For Tier 1:

$$\Delta \text{term\_Ubi} = 0.5 \cdot ug1\_base \cdot \text{sSFR} \cdot e^{-t/\tau_{\text{SF}}}$$

For Tiers 2 and 3 (at  $t=0$ ):

$$\Delta \text{term\_F\_UBii}|_{t=0} = -\beta_i \cdot ug1\_base \cdot \text{sSFR} \cdot \omega_g \cdot \frac{M_0}{r} \cdot [UA]$$

$$\Delta \text{term\_Ub\_i}|_{t=0} = -\beta_i \cdot ug1\_base \cdot \text{sSFR} \cdot \omega_g \cdot \frac{M_{\text{Fornax}}}{r_{\text{Fornax}}} \cdot [UA]$$

The total coherent buoyancy boost is:

$$\Delta g_{\text{buoy\_total}} = \text{sSFR} \cdot (\text{term\_Ubi}^\infty + \text{term\_F\_UBii}^\infty + \text{term\_Ub\_i}^\infty) \cdot e^{-t/\tau_{\text{SF}}}$$

where the superscript  $\hat{\infty}$  denotes the static (non-sSFR) component amplitudes.

#### 3.2 Key Prediction

**Starburst-buoyancy coherence:** The peak buoyancy enhancement  $\hat{\Delta} g_{\text{buoy\_total}}(t=0)$  occurs simultaneously with peak sSFR. Both decay with the **same timescale**  $\hat{\tau}_{\text{SF}} = 100 \text{ Myr} = 3.15576 \times 10^{14} \mu \text{ s}$ . This is a unique prediction: in standard Newtonian gravity, buoyancy has no dependence on star formation rate.

#### 3.3 Numerical Values for NGC 1792

| Parameter                              | Value                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------|
| sSFR                                   | $10^{-11} \text{ yr}^{-1}$                                                  |
| $ug1\_base$                            | $G \hat{M} / r^2 \approx 7.35 \times 10^{-11} \text{ m/s}^2$                |
| $\hat{\Gamma}_{\text{Tier 1 at } t=0}$ | $0.5 \times 7.35 \times 10^{-11} \approx 3.7 \times 10^{-12} \text{ m/s}^2$ |
| $\hat{\tau}_{\text{SF}}$               | $100 \text{ Myr} = 3.15576 \times 10^{14} \mu \text{ s}$                    |
| Coherence decay time                   | 100 Myr (matches SF episode)                                                |
| $\hat{\Gamma}_i$                       | 0.61                                                                        |
| $\hat{\Gamma}_g$                       | $7.3 \times 10^{-11} \text{ rad/s}$                                         |

The coherence ratio (buoyancy enhancement / static buoyancy) at  $t=0$ :

$$\mathcal{C}_{\text{NGC1792}} = \frac{\Delta g_{\text{buoy\_total}}(0)}{g_{\text{buoy\_static}}} = \text{sSFR} \approx 10^{-9} \text{ yr}^{-1}$$

This is an intrinsic signature of the specific star-formation rate being encoded in the gravitational buoyancy field.

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## 4. Observable Signatures

### 4.1 Enhanced Gravitational Buoyancy at Starburst Epoch

Galaxies with high sSFR ( $\text{sSFR} > 10^{-9} \text{ yr}^{-1}$ , characteristic of starburst galaxies) should show measurably enhanced gravitational buoyancy across all 3 UQFF tiers simultaneously. The coherence ratio  $\mathcal{C}$  scales linearly with sSFR.

### 4.2 sSFR-Buoyancy Correlation

The UQFF prediction is: galaxies with high sSFR should exhibit: - Enhanced Tier 1 (static half-gravity) during starburst - Enhanced Tier 2 (dynamic compact oscillatory) with same 100 Myr timescale - Enhanced Tier 3 (outer-frame coupling to Fornax environment) scaled by same factor

This creates a **universal sSFR-buoyancy scaling relation**:

$$g_{\text{buoy\_enhanced}}(\text{sSFR}) = g_{\text{buoy\_passive}} \times (1 + \text{sSFR} \times \tau_{\text{obs}})$$

### 4.3 Starburst Quenching Imprint

When star formation is quenched ( $\dot{M}_{\text{SF}} \hat{=} 0$ ), the buoyancy enhancement drops to zero on the SF timescale. This should be observable as correlated suppression of gravitational signatures alongside AGN-quenching or SN-driven gas expulsion.

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## 5. Astrophysical Context

NGC 1792 has a well-measured SFR from infrared and  $\text{H I} \pm$  studies. The normalized  $\text{sSFR} = 10 \text{ M}_{\odot}/\text{yr} / 10^{11} \text{ M}_{\odot} = 10^{-9} \text{ yr}^{-1}$  is characteristic of actively star-forming disk galaxies. The coupling of sSFR to the UQFF buoyancy field via  $M(t)$  is a natural consequence of the PAPER\_198 3-tier framework when applied to time-evolving mass systems.

The Fornax Cluster ( $M_{\text{Fornax}} = 7 \times 10^{14} \text{ M}_{\odot}$ ,  $r_{\text{Fornax}} \hat{=} 20 \text{ Mpc}$ ) as the Tier 3 external body introduces the large-scale gravitational environment. The outer-frame coupling term  $U_{\text{b}_i}$  carries information about the cluster environment into the local gravitational field of NGC 1792, weighted by sSFR.

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## 6. Conclusions

1. In the NGC 1792 UQFF 2.0 model, SFR\_factor is the **specific star-formation rate** ( $\text{sSFR} = 10^{-11} \text{ yr}^{-1}$ ), which acts as a **dimensionless coupling constant** modulating all three buoyancy tiers.
2. The resulting **starburst-buoyancy coherence** prediction: all three UQFF buoyancy tiers peak simultaneously with star formation and decay with the same 100 Myr timescale.
3. The coherence formula is:  $\dot{g}_{\text{buoy\_total}} = \text{sSFR} \times (U_{\text{bi}} + F_{\text{UBii}} + U_{\text{b\_i}}) \times e^{\{t/\tau_{\text{SF}}\}}$ .
4. The coherence ratio  $C = \text{sSFR} \times 10^{-11} \text{ yr}^{-1}$  for NGC 1792, providing a direct observational link between the galaxy's star-formation rate and its gravitational buoyancy signature.
5. This predicts a universal  $\text{sSFR} \propto \text{buoyancy}$  scaling relation testable across the galaxy population.

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**UQFF computed:** UQFF magnetic Jeans correction factor  $[SSq] \times B^2 / (8\pi \times \rho \times c_s^2) = 5.7\text{e-}1 \times 1.3\text{e-}9 = 7.4\text{e-}10$ ; Jeans mass deviation from standard  $= 7.4\text{e-}10 \times M_J$ .

## References

- Daniel T. Murphy, *UQFF Framework*, Star-Magic Repository (2025)
- PAPER\_198: F\_UBii Taxonomy Part 1 – Compact Object and Stellar Buoyancy
- GALAXY\_NGC\_1792.cpp UQFF 2.0 (Session 73, Module 19)
- NGC 1792 observational parameters: HYPERLEDA / NED database
- Fornax Cluster parameters: Drinkwater et al. (2001),  $M_{500} = 7 \times 10^{13} M_\odot$

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